Exp. 8 Uploading Architecture Diagram (MVC & Microservices) in Azure DevOps Wiki

DEFINITION:

An Architecture Diagram represents the overall structure of a system, showing components, their relationships, and how they interact.
MVC (Model-View-Controller) and Microservices architectures are commonly used to design scalable and maintainable applications.

PURPOSE:

• To represent system architecture clearly

• To show separation of concerns using MVC

• To illustrate distributed system design using Microservices

• To understand component interaction and data flow

BASIC COMPONENTS (Notations):

1. Model (MVC)

• Represents data and business logic

• Handles database interactions

Example: User Model, Product Model

2. View (MVC)

• Represents the UI layer

• Displays data to users

Example: Web page, Mobile UI

3. Controller (MVC)

• Handles user input and requests

• Connects Model and View

Example: Login Controller

(MVC & Microservices) in Azure DevOps Wiki

DEFINITION:

An Architecture Diagram represents the overall structure of a system, showing components, their relationships, and how they interact.
MVC (Model-View-Controller) and Microservices architectures are commonly used to design scalable and maintainable applications.

PURPOSE:

• To represent system architecture clearly

• To show separation of concerns using MVC

• To illustrate distributed system design using Microservices

• To understand component interaction and data flow

BASIC COMPONENTS (Notations):

1. Model (MVC)

• Represents data and business logic

• Handles database interactions

Example: User Model, Product Model

2. View (MVC)

• Represents the UI layer

• Displays data to users

Example: Web page, Mobile UI

3. Controller (MVC)

• Handles user input and requests

• Connects Model and View

Example: Login Controller

4. Microservices

• Independent services performing specific tasks

• Communicate via APIs

Example: Auth Service, Payment Service

5. API Gateway

• Entry point for all client requests

• Routes requests to appropriate services

6. Database

• Stores data for services

• Can be separate for each microservice

7. Communication

• Interaction between services

• REST APIs / HTTP / Messaging

HOW TO UPLOAD ARCHITECTURE DIAGRAM

AIM:

To upload an Architecture Diagram (MVC & Microservices) in the Wiki using Azure DevOps.

TOOLS REQUIRED:

• Azure DevOps

• Web browser

• Architecture Diagram image (PNG/JPG)

PROCEDURE:

Step 1: Open Azure DevOps

• Go to Azure DevOps

• Sign in with your Microsoft account

Step 2: Select Project

• Choose your existing project from the dashboard

Step 3: Navigate to Wiki

• Click on Wiki from the left menu

• If Wiki is not created:

o Click Create Wiki

o Select Project Wiki

Step 4: Create New Page

• Click on New Page

• Enter Architecture Diagram

Step 5: Upload Diagram

• Click on Insert Image (image icon)

• Choose Upload from Computer

• Select the Architecture Diagram file

• Click Upload

Step 6: Save the Page

• Click Save

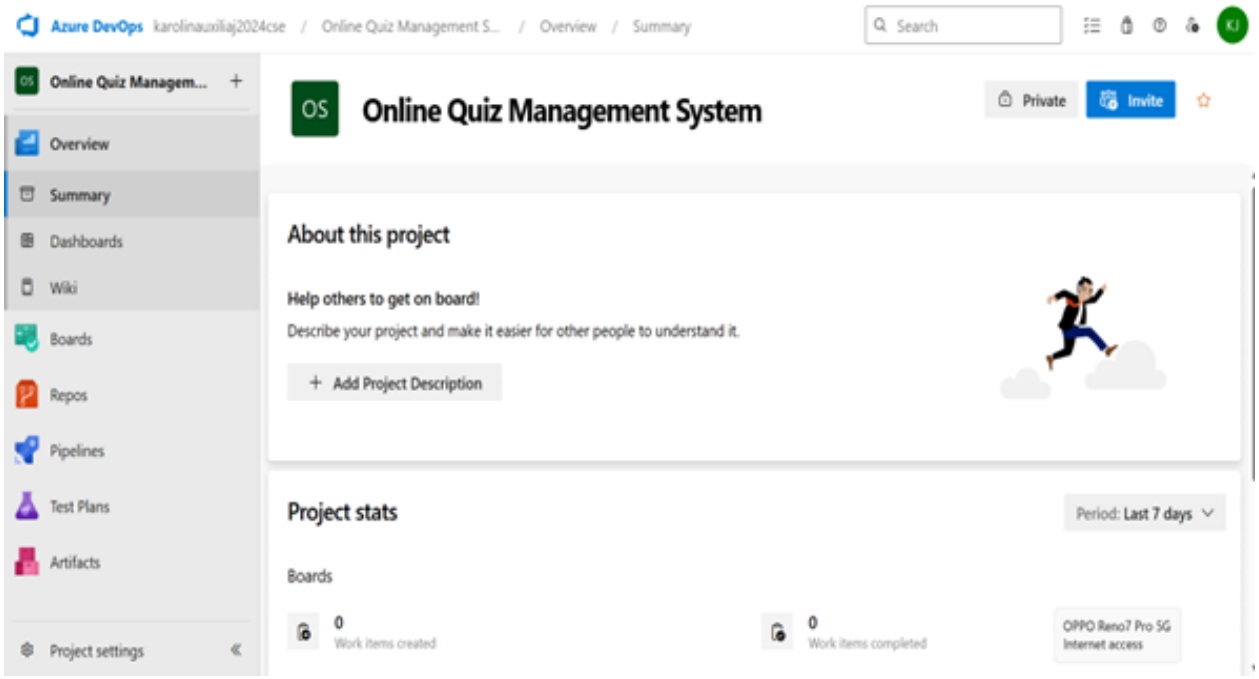
Step 7: Verify Upload

• Ensure the Architecture Diagram is displayed correctly

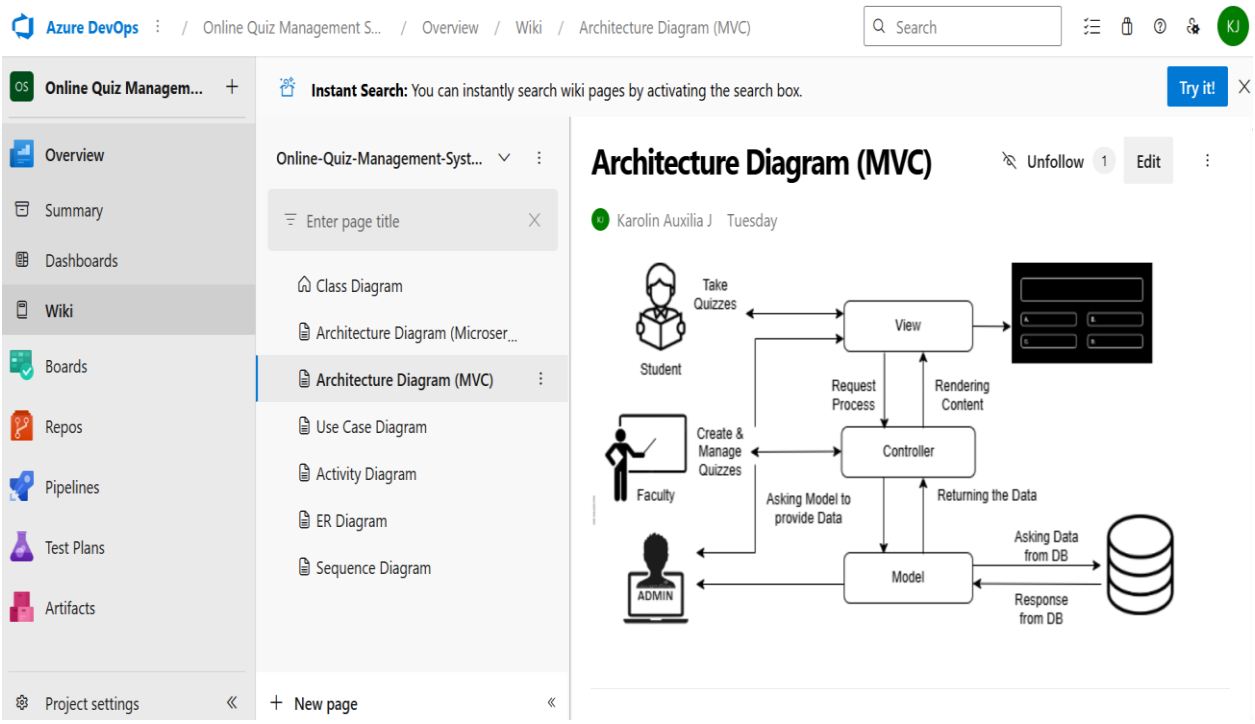
RESULT:

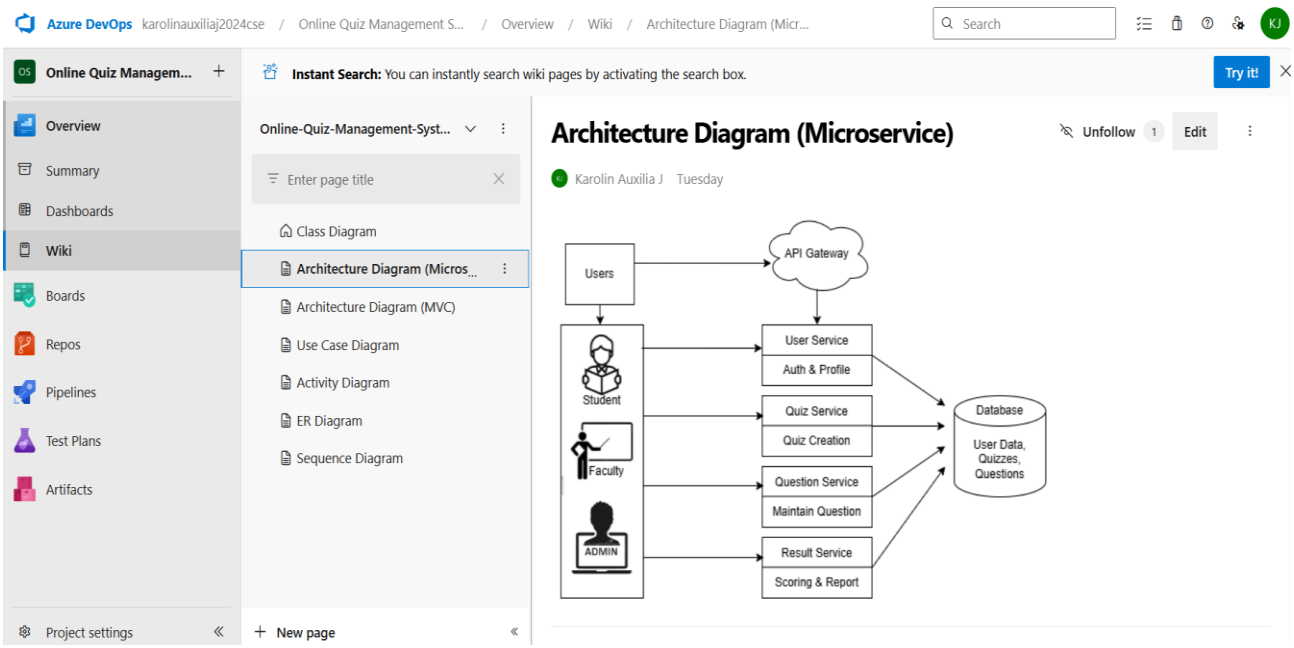
Thus, the Architecture Diagram using MVC and Microservices was successfully uploaded to the Wiki using Azure DevOps.

Project Page

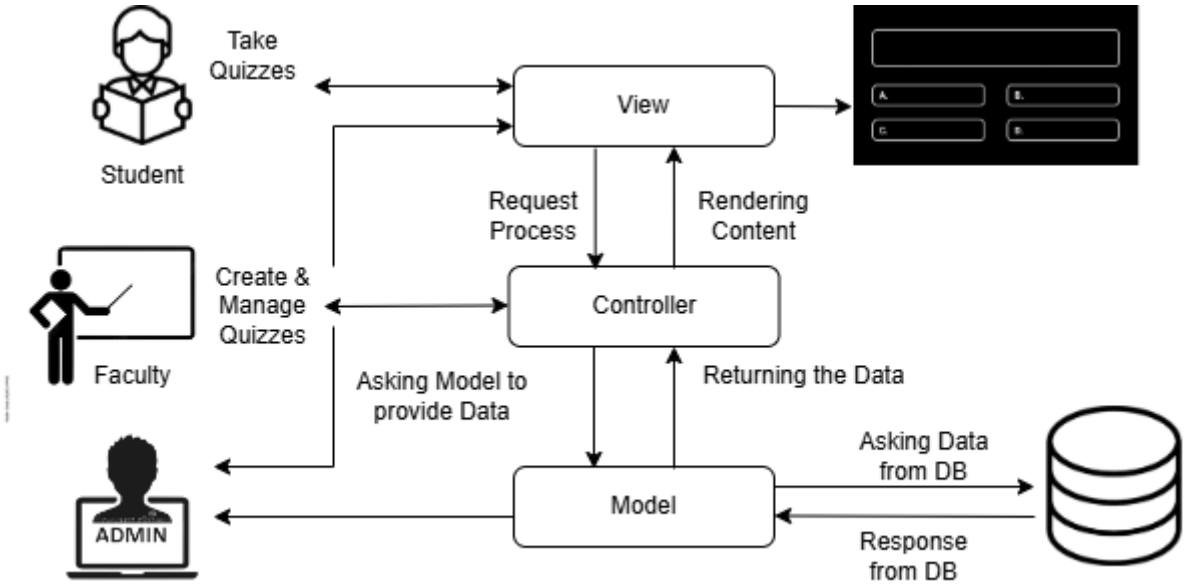
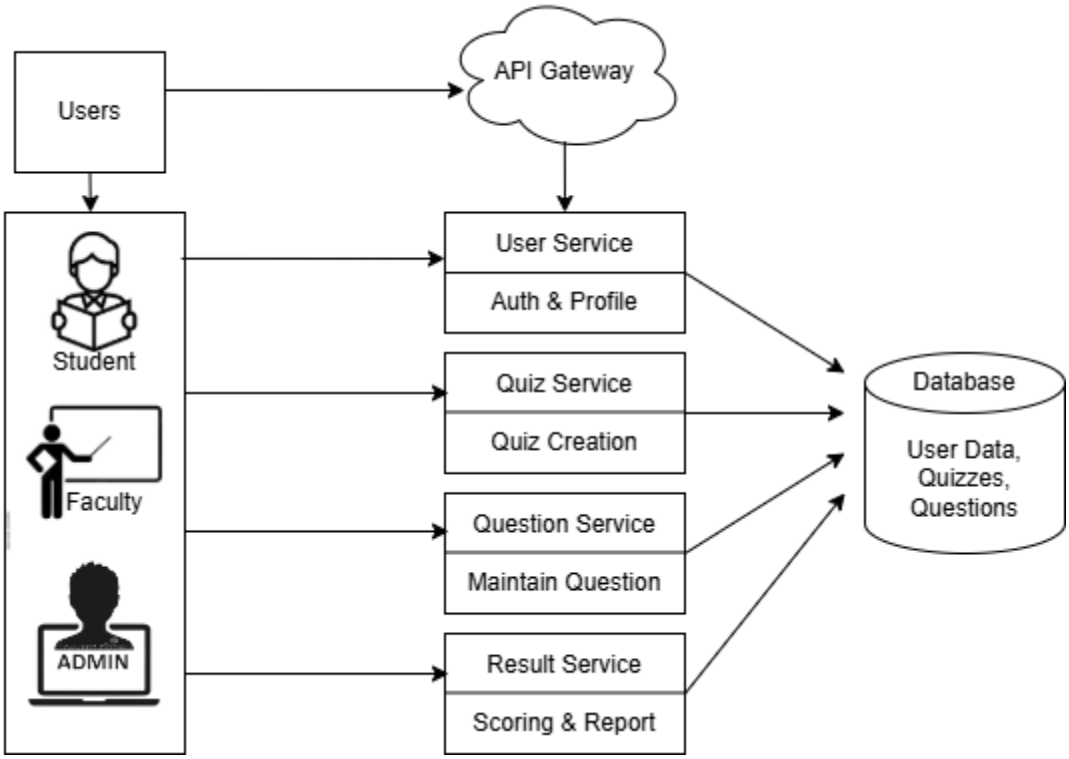


Wiki Page





MVC Architecture Diagram



Microservice Architecture Diagram